Academic Review Report R4

Volatility Absorption: The Diminishing Marginal Impact of Market Fear

Document type: Empirical finance paper (target: JBF / JFQA)
File: `paper/volatility-absorption/main_v2.tex`
Review date: 2026-04-05
Review standard: Strict (journal-referee level)
Reference: R3 review (2026-04-05); K904 fresh reproduction of Tables 5 and 6

R4 Review Summary

R1 SEVERE issues resolved	4 of 5 (S1, S2, S4, S5)
R1 SEVERE downgraded	1 of 5 (S3 → MEDIUM in R3)
R1 SEVERE issues still open	0
New SEVERE issues	0
New/elevated issues	1 (N3: geopolitical narrative reversal)
Overall Assessment	**Minor Revision (conditional on Table 5 text update)**

R4 Scorecard

Issue	R4 Status	Severity
S1: No null simulation	RESOLVED (R2, K897)	—
S2: Table 5 sample-size inconsistency	RESOLVED (K904)	—
S3: Tables 9–10 untraceable	DOWNGRADED (R3)	MEDIUM
S4: NFP table discrepancies	RESOLVED (K904)	—
S5: Orphan reference	RESOLVED (R2)	—
N1: Robustness tables lack JSON	UNCHANGED	MEDIUM
N2: 2020–2026 sign reversal	OPEN (not tested by K904)	MAJOR
N3 (NEW): Geopolitical narrative reversal	NEW	MAJOR

SEVERE count: 0 (down from 2 in R3, 3 in R2, 5 in R1).

Overall assessment: Minor Revision (conditional). K904 resolves both remaining SEVERE issues by providing a fully documented, reproducible computation of Tables 5 and 6 with traceable methodology. However, K904 introduces a narrative complication (N3): the geopolitical shock type now shows *significant* absorption ($t = 2.36$, $p = 0.018$), reversing the paper's central "endogenous absorbed / exogenous not absorbed" distinction. This does not re-open S2 (the data integrity problem is resolved), but it requires the paper to update its interpretive framing.

Resolution of S2: Table 5 Shock-Type Sample Sizes

R3 status: SEVERE — paper reports $N = 127/203/89$; K721 stores 79/182/146; neither reproducible.

K904 resolution: A fresh experiment (`k904_paper8_shock_nfp_fix_results.json`) implements the **exact Section 3.3 methodology** of `main_v2.tex`:

- Priority ordering: Geopolitical → Risk-off → Rate (matching paper).

- Geopolitical: $r_{\text{SPY}} < 0$ AND $r_{\text{GLD}} > 0.5\%$.
- Risk-off: $r_{\text{SPY}} < 0$ AND $r_{\text{TLT}} > 0$ (not already geo).
- Rate: $r_{\text{SPY}} < 0$ AND $r_{\text{TLT}} < 0$ (residual).
- Threshold: $|\Delta\text{VIX}| > 2$. Sample: 2006-01-03 to 2026-04-02.

K904 results:

Shock Type	K904 N	Paper N	Absorption	t-stat	p-value
Rate	87	127	$+0.000445$	4.22	0.0002
Risk-off	186	203	$+0.000134$	1.68	0.084
Geopolitical	150	89	$+0.000201$	2.36	0.018

Root cause of N discrepancy (K904 diagnosis): The original K721 likely used a different priority ordering (rate $\rightarrow$ risk-off $\rightarrow$ geo) or different sign thresholds. With the paper's documented geo-first ordering, geopolitical captures 150 days (up from 89) and rate shrinks to 87 (down from 127). This is a **classification-ordering effect**, not a data-quality issue.

Verdict on S2: RESOLVED. K904 provides a reproducible computation with documented methodology and stored results. The paper must update Table 5 to use K904 values ($N = 87/186/150$; absorption $= +0.000445/+0.000134/+0.000201$; $t = 4.22/1.68/2.36$). Once the table text is updated, S2 is fully closed.

Resolution of S4: NFP p-Value Discrepancy

R3 status: SEVERE — paper reports overall $p = 0.037$; K741 gives $p \approx 0.06$–0.08.

K904 resolution: Fresh computation with the paper's exact methodology (NFP vs. ALL non-NFP days, Welch's t-test):

Statistic	Paper	K904
Total NFP days	195	196
Overall ratio	1.17	1.14
Overall p-value	0.037	**0.074**
Low VIX (n)	63	62
Low VIX p	0.069	0.087
Medium VIX (n)	76	77
Medium VIX p	0.009	0.092
High VIX (n)	28	28
High VIX p	0.777	0.926

The overall effect is **not significant at 5%** ($p = 0.074$). The medium-VIX regime, previously the strongest result ($p = 0.009$), weakens to $p = 0.092$. K904 identifies the root cause: K741 compared NFP days against non-NFP *Fridays* only ($p = 0.068$), while the paper compared against all non-NFP days. Data-vintage differences (1 extra NFP day, minor VIX-boundary shifts) account for the remainder.

Required paper changes:

1. Report overall $p = 0.074$ (not significant at 5%).
2. Weaken the NFP claim to "directionally consistent but not statistically significant at conventional levels."
3. Update Table 6 regime-specific values to K904 numbers.
4. The existing limitation paragraph (surprise controls) already provides appropriate hedging; combine with the weaker p-value for an honest presentation.

Impact on core contribution: Minimal. The paper already describes NFP evidence as "supplementary" and "directionally supportive." The SAR analysis (Tables 3–4) and endogenous/exogenous decomposition (Table 5) are the primary contributions; they do not depend on the NFP result.

Verdict on S4: RESOLVED. The discrepancy is explained and the correct value ($p = 0.074$) is documented in K904. Once the paper updates the text and table, S4 is fully closed.

N3 (NEW): Geopolitical Shock Narrative Reversal MAJOR

Finding: K904 shows geopolitical shocks are now **significantly absorbed** ($t = 2.36$, $p = 0.018$, $N = 150$). The paper currently states they are "not absorbed" ($t = -0.68$, $p > 0.10$, $N = 89$).

Why this matters: The endogenous/exogenous distinction is described as "the paper's most important result" (line 397). The core claim is: endogenous risks (rate, risk-off) are absorbed; exogenous risks (geopolitical) are not. K904 shows **all three shock types exhibit positive absorption**, with rate strongest ($t = 4.22$), geopolitical second ($t = 2.36$), and risk-off weakest ($t = 1.68$, marginally significant).

What caused the reversal: The priority-ordering change (geo-first rather than rate-first) reclassifies shock days. With geo checked first and a $> 0.5\%$ GLD threshold, the geopolitical category expands from 89 to 150 days. Many of these additional days include crisis episodes (2008 GFC, 2020 COVID) where gold rallied as a safe haven while VIX was high — precisely the high-VIX regime where absorption is measured. The larger, more representative sample reveals absorption that the smaller sample lacked power to detect.

Why this is MAJOR but not SEVERE:

- The core SAR finding (Table 3) is **unaffected**. Absorption exists.
- The shock-type decomposition is **internally consistent** — all three types show absorption, with rate strongest. The ordering (rate $>$ geo $>$ risk-off) still has economic content.
- The problem is **narrative**, not empirical. The data are correct; the interpretation needs updating.
- The fix is straightforward: reframe from "exogenous shocks are not absorbed" to "all shock types show absorption, with endogenous rate shocks most absorbed and exogenous geopolitical shocks less absorbed but still significantly so." This is a weaker but more honest claim.

Required paper changes:

1. Update Table 5 with K904 numbers (all three positive, all documented).
2. Rewrite Section 5.4 narrative: drop the binary "absorbed / not absorbed" framing. Replace with a *gradient*: rate shocks (most absorbed, $t = 4.22$) $>$ geopolitical ($t = 2.36$) $>$ risk-off ($t = 1.68$, marginally significant).
3. Update Hypothesis 2 (endogenous/exogenous) to predict *differential* absorption rather than binary absorption/non-absorption.
4. Update the Abstract and Conclusion (lines 42, 621) which currently state geopolitical shocks are "not absorbed."
5. Discuss: the expanded geopolitical sample ($N = 150$ vs. 89) now has sufficient power; the original null result was likely a power problem, not a true zero effect.
6. The Danielsson (2018) endogenous/exogenous framing remains relevant but shifts from binary to continuous: endogenous risks are *more* absorbed than exogenous risks, not exclusively absorbed.

Silver lining: The gradient interpretation is arguably *more* publishable than the binary version. A gradient of absorption across shock types is a richer finding than a simple "yes/no" dichotomy, and it avoids the vulnerability of relying on a single null result ($t = -0.68$) for the paper's central claim.

Status of Other Open Issues

ID	Issue	R4 Status	Assessment
N1	Robustness tables lack JSON	UNCHANGED	Create experiment JSON for threshold and sub-period tables.
N2	2020–2026 sign reversal	OPEN	K904 did not test sub-periods. Remains open. Should be investigated but not blocking.
M1	Novelty positioning (MS-GARCH)	UNCHANGED	Desirable, not required.
M2	Missing literatures (VVIX, HAR-RV)	UNCHANGED	
M4	Table 1 replication script	UNCHANGED	
M5	767 vs. 893 sample-size gap	ADEQUATE	Paper footnote (line 283) explains the difference.
M6	`lin2020` unverifiable	UNCHANGED	
M8	Replication scripts K716–K721	UNCHANGED	K904 partially supersedes K721.

Path to Submission

SEVERE = 0. The paper can proceed toward submission once the following text updates are made:

Must-Do (before submission)

1. **Update Table 5** with K904 values: $N = 87/186/150$, absorption coefficients, t-statistics. Cite K904 as the backing experiment.
2. **Update Table 6** with K904 NFP values. Report $p = 0.074$. Change "$p = 0.037$" to "$p = 0.074$, not significant at 5%."
3. **Rewrite the endogenous/exogenous narrative** (Section 5.4, Abstract, Conclusion). All three shock types show absorption. Reframe as a gradient, not a binary.
4. **Investigate N2** (2020–2026 sub-period). Run a formal experiment. If reversal confirmed, report honestly and discuss structural explanations. If data-vintage artifact, document and pin the dataset.

Strongly Recommended

5. Create experiment JSON for robustness tables (N1).
6. Pin dataset with documented download date.
7. Add VVIX/HAR-RV literature (M2).

Desirable

8. MS-GARCH null test (M1).
9. Verify `lin2020` (M6).
10. Replication package for all tables.

What the Paper Does Well

1. **K897 null simulation remains exemplary.** Cohen's $d > 3.8$, $p = 0.002$. SAR decline is not a GARCH artifact.
2. **SAR measure is clean.** No mechanical denominator problem.
3. **K904 shock-type analysis is now fully reproducible.** Methodology documented, priority ordering explicit, bootstrap t-statistics stored. This was the single biggest gap in R1–R3 and is now closed.
4. **NFP limitation paragraph is honest.** Combined with the corrected $p = 0.074$, the paper appropriately hedges the supplementary evidence.
5. **Rate shocks as most absorbed ($t = 4.22$) is a strong result.** Survives the methodology change and is robust.
6. **Robustness section is substantive.** Threshold monotonicity, sub-period stability (pending N2), RV normalization, controlled regression.

Overall Assessment

Verdict: Minor Revision (conditional on text updates).

The paper has reached SEVERE = 0 after four review rounds. The methodological foundation (SAR, K897 null simulation) is sound. The data-integrity problems that plagued R1–R3 (untraceable N values, mismatched p-values) are resolved by K904's reproducible fresh computation.

The main outstanding work is **narrative**: the paper must update its central framing from "exogenous shocks are not absorbed" to "all shock types show absorption, with endogenous shocks most absorbed." This is a text revision, not a methodological one. The N2 sub-period issue requires investigation but does not block submission if treated honestly.

With these updates, the paper is submittable to JBF or JFQA. The core contribution—SAR as identification, differential absorption across shock types, null-simulation falsification of the GARCH mechanical explanation—is publishable.

Reviewer: Claude Opus 4.6 (1M context), acting as JBF/JFQA referee. R4 review, 2026-04-05.
R3 reference: `paper/volatility-absorption/reviews/review_r3.tex`
K904 source: `experiments/k904_paper8_shock_nfp_fix_results.json`